Module 5: Pooling Layers

1. What is pooling?

Pooling is a downsampling operation commonly used in CNNs to reduce the spatial dimensions of feature maps.

It summarizes values in small regions using a rule such as maximum or average.

Pooling is separate from convolution.

2. Max pooling

Max pooling selects the largest value from each region.

Suppose the input feature map is:

1 3 2 4

5 6 1 2

7 2 8 9

4 1 3 5

Using a 2 × 2 window and stride 2:

Top-left region:

1 3

5 6

Maximum = 6

Top-right region:

2 4

1 2

Maximum = 4

Bottom-left region:

7 2

4 1

Maximum = 7

Bottom-right region:

8 9

3 5

Maximum = 9

Output:

6 4

7 9

Max pooling retains the strongest activation in each region.

3. Average pooling

Average pooling calculates the average value in each region.

For the top-left region:

1 3

5 6

Average = (1 + 3 + 5 + 6) / 4

Average = 3.75

The output represents the average activation rather than the maximum.

4. Min pooling

Min pooling selects the smallest value in each region.

It is possible to use min pooling, but it is much less common in standard CNN architectures than max pooling or average pooling.

5. Why do we use pooling?

Pooling can:

- Reduce feature-map height and width.
- Reduce the computation required by later layers.
- Summarize local feature responses.
- Provide some tolerance to small changes in feature position.

However, pooling can also discard useful spatial information.

Some CNN architectures use strided convolution instead of pooling.

6. Does pooling reduce channels?

Ordinary pooling is applied independently to each channel.

For example:

Before pooling: $32 \times 32 \times 16$

After 2×2 pooling with stride 2:

$16 \times 16 \times 16$

The height and width are reduced, but the number of channels remains 16.

Conclusion

Max pooling selects the maximum, average pooling calculates the mean, and min pooling selects the minimum. Pooling generally reduces spatial dimensions while preserving the channel count.